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	Q.1	

a)) Given Message : Rjjy rj ts ymj xfggfym,
Bj bnqq inxhzxx ymj
Uqfs

→ Decoding each word

R j j y
↓ ↓ ↓ ↓
M e e t

b n q q
↓ ↓ ↓ ↓
y k m m

→ rj
↓ ↓
o g

i n x h z x x
↓ ↓ ↓ ↓ ↓ ↓ ↓
f k u e w u u

→ ts
↓ ↓
q p

y m j
↓ ↓ ↓
v j g

→ y m j
↓ ↓ ↓
v j g

u q f s
↓ ↓ ↓ ↓
r n c p

→ x f g g f y m
↓ ↓ ↓ ↓ ↓ ↓ ↓
a c d d c v l

→ B j
↓ ↓
w e

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		Putting all together "Meet og qp vjg ucddcvl. We ykmm fkuewuu vjg xxcp"
	b))	1. Represent the P_T message numerically R → 82 I → 73 S → 83 K → 75 2. Encryption using public key (e, n) encryption formula is $C = M^e \bmod n$ where, M is P_T $e = 5$ $n = 35$ Perform encryption for each value in P_T 1. $C_1 = 82^5 \bmod 35 = 17$ $C_2 = 73^5 \bmod 35 = 33$ $C_3 = 83^5 \bmod 35 = 13$ $C_4 = 75^5 \bmod 35 = 10$ 3. Decryption using Private key d

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Decryption formula is

$$M = c^d \bmod n$$

where,

c = ciphertext
$d = 29$
$n = 35$

$\therefore P_1 = 17^{29} \bmod 35 = 12$
$P_2 = 33^{29} \bmod 35 = 3$
$P_3 = 13^{29} \bmod 35 = 13$
$P_4 = 10^{29} \bmod 35 = 5$

The decrypted message does not match the original P_T due to the limitations of $n = 35$ which is smaller than numerical representation of some P_T letters.

c) Steganography is the practice of concealing information within another medium such as images, audio files or text in a way that the hidden message remains undetectable to a casual observer.

The difference between steganography & encryption:

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Aspect	Steganography	Encryption
Purpose	Hides the Presence of a message	Makes a message Unreadable without a key
Visibility	Message remains invisible	encrypted data is obvious
Key dependancy	May or may not require key	Requires a key for encryption & decryption
Security approach	Relies on Secrecy	Relies on the computation difficulty

~~Application of Steganography~~ :-

1. Covert Communication
2. Digital watermarking
3. Data Integrity Verification
4. Military & intelligence operations

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	Q.2	
	a))	The diffie-hellman key exchange involves following Calculations:

Given:

1) Public Prime number $P = 23$

2) Primitive root $g = 5$

3) Alice's Private key $a = 6$

4) Bob's Private key $b = 15$

S1: Calculate Alice Public key A

$$A = g^a \bmod P$$

$$A = 5^6 \bmod 23$$

S2: Calculate Bob's Public key B

$$B = g^b \bmod P$$

$$= 5^{15} \bmod 23$$

S3: Compute the shared secret key

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$S = B^a \bmod P$

2. Bob's Calculah

$S = A^b \bmod P$

The values for S will match if Calculations are Correct

∴ Alice Public key (A)

$A = 8$

Bob's Public key (B)

$B = 19$

Shared Secret key (S)

$S = 2$

∴ Alice Computes $S = B^a \bmod P$

$= 19^6 \bmod 23 = 2$

Bob Computes $S = A^b \bmod P$

$= 8^{15} \bmod 23 = 2$

∴ $S = 2$

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	b))	SSL / TLS (Secure Sockets Layer / Transport Layer Security)

SSL/TLS ensures a Secure Connection between email clients & Servers or between two email Servers by encrypting the Communication channel. This Prevents attackers from intercepting or altering email Content during Transmission It Protects emails from eavesdropping while they travel over the internet. It uses Certificates to verify the identity of email Servers ensuring users are Communicating with trusted Servers SSL/TLS Protects emails only during transit, once the email reaches the receipts Server or inbox, it is no longer encrypted.

S/MIME provides end-to-end encryption for email Content & ensures the authenticity of the Sender through digital Signatures It uses public Key Cryptography to encrypt the emails Content, ensuring only the intended receipt can decrypt & read it. It also Verifies the Sender's identity & ensures that the message was not altered after it was Sent. Supported natively by many email clients

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PGP is a widely used standard for securing emails, providing both encryption & digital signatures. It uses a decentralized model of trust unlike S/MIME certificate authority based model. It ensures that only the receipent can decrypt the email content using their private key. It confirms the sender's identity & guarantees the integrity of the email. It allows users to verify other public keys directly offering as alternative to centralized certificate authorities

Comparison ← Contribution to Email Security

Protocol	Encryption	Authentication	Scope
SSL/TLS	encrypts comm channel	Verifies Servers	during tx only
S/MIME	end-to-end encryption	digital signatures	full message security
PGP	end to end encryption	Digital signature	full message security

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	Q. 3	

a)) S1 :- Construct the 5×5 Key matrix.
1. Write the keyword "MONARCHY" without repeating any letter
2 - Include I/J in same block

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

S2. Divide P_T into diagraphs

BA — IB
LX — PS
LO — PM
ON — NA

∴ the encrypted text is IBPSPMNA

b)) System Corruption refers to malfunction or breakdown of a Computer System's functionality due to alteration or damage of its operating System, files or Configuration. It can render the System Partially or Completely Unusable, often leading to data loss, Performance degradation or

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Compromised Security

Malicious Software a malware is a Primary cause of System Corruption, with intent to harm, exploit & gain Unauthorized access to it.

Malware achieves System Corruption by

1. Injecting malicious code
2. Deleting or encrypting critical System files
3. Overwriting System Configurations
4. Spreading across Connected System

Examples of malware that Cause System Corruption

1. Ransomware : It encrypts the Victim data or critical System files, rendering them inaccessible until a ransom is paid.

2. Trojan Horse : Trojans disguise themselves as legitimate Programs but Once executed introduce harmful Code that Can delete or modify System files

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		System corruption through malicious Poses a significant threat to individuals & organizations
	Q.4	
	a)	The DES is a symmetric key block cipher that operates on 64 bit blocks of P_T & encrypts them into 64 bit C_T using a 56-bit key. DES follows a feistal n/w structure dividing the P_T into two halves & Performing iterative rounds of Substitution & Permutation
		Main Steps in encrypting a 64-bit Block using DES
		1. Initial Permutation: Rearranges the Plain text in a Predefined table
		2. Split into halves: The block is divided into two 32-bit halves the left half & right half
		3. 16 round of feistal operations: for each round i (1 to 16).
		1- Expansion
		2. Key mixing

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5. Substitution
6. Permutation
5. mixing

The new Li & Ri are Passed to nexet Round

4. Final Permutation

DES encrypts a 64-bit block using a feistal n/w with 16 rounds of Processing

The 16 Subkeys are derived from a 56-bit key using Permutation, Splitting, Cyclic Shifting & Compression.

b)) SNMPV3 is designed to enhance the Security of n/w management operations by addressing the limitations of earlier Versions. It introduces robust mechanism for authentication & encryption ensuring the Confidentiality, integrity & authenticity of the data exchanged between n/w management Systems & agents.

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SNMPV3 employs Security models & Security levels to provide flexible & layered Protection for management information the Primary Security model is the user based Security model (USM) with incorporates authentication & encryption

SNMPV defines 3 Security levels that determine the Combinations of authentication & encryption used :

1. NoAuthNoPriv
2. authNoPriv
3. authPriv

Ensuring Confidentiality & integrity in SNMPV3

1. Confidentiality
2. Integrity
3. Authentication
4. Replay Protection